BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

MATHEMATICS

Past Paper 2024 • Paper Code: PR–XI (A-I-B) (24)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 20 | **Paper Code:** PR–XI (A-I-B) (24)

ا۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. ${}^{n}C_{n-1}$ = _____.

(A) n
(B) n−1
(C) n+1
(D) n!

Q2. In the expansion of $(a+b)^5$ the Middle term is _____.

(A) 3rd
(B) 4th
(C) 2nd and 3rd
(D) 3rd and 4th

Q3. Which of the given is even function?

(A) X^2
(B) $2x^4$
(C) $2x^4-x^2+3$
(D) All these

Q4. Domain of $f(x) = x/(x^2-16)$ is _____.

(A) $R-\{-4,4\}$
(B) R
(C) $\{-4,4\}$
(D) $\{0,4,-4\}$

Q5. $\cos(\pi+\theta)$ = _____.

(A) $\cos\theta$
(B) $\sin\theta$
(C) $-\cos\theta$
(D) $-\sin\theta$

Q6. $2\sin\alpha\cos\beta$ = _____.

(A) $\sin(\alpha+\beta)+\sin(\alpha-\beta)$ (B) $\sin(\alpha+\beta)-\cos(\alpha+\beta)$

(C) $\sin(\alpha+\beta)-\sin(\alpha-\beta)$ (D) $\sin(\alpha+\beta)+\cos(\alpha+\beta)$

Q7. If a,b,c are the sides of triangle ABC then a = _____.

(A) $b\sin\alpha/\sin\beta$ (B) $\sin\alpha/b\sin\beta$

(C) $b\sin\beta/\sin\alpha$ (D) None of these

Q8. The circum radius of a triangle ABC having sides a,b,c is given by _____.

(A) r = /s (B) R = abc/4

(C) R = /(s−a) (D) R = 4/abc

Q9. Period of (1/2)tan3x is _____.

(A) 2π (B) 6π

(C) π/3 (D) π/6

Q10. If f(−x) = −f(x) for every 'x' in the domain of f. Then f is called _____.

(A) Odd function (B) Even function

(C) Constant function (D) Identity function

Q11. IF Z is any complex number then |−Z| = _____.

(A) $\sqrt{(-a^2-b^2)}$ (B) $\sqrt{(a^2-b^2)}$

(C) $\sqrt{(a^2+b^2)}$ (D) $\sqrt{(b^2-a^2)}$

Q12. i^{-9} = _____.

(A) 1/i (B) ±i

(C) −1/i (D) i

Q13. If A = [[1,2,3],[6,5,4],[7,8,9]] then the minor M13 = _____.

(A) 13 (B) −13

(C) −12 (D) 36

Q14. If 'A' is a square matrix of order 3, then $A+A^t$ is _____.

(A) Symmetric (B) Skew symmetric

(C) Zero matrix (D) Hermition matrix

Q15. If u = 3i − 2j then û = _____.

(A) √13 (B) √13(3i−2j)

(C) 13 (D) (1/√13)(3i−2j)

Q16. If V = xi + yi + Zk then |V| = _____.

(A) $\sqrt{(x^2+y^2+z^2)}$ (B) $x^2+y^2+z^2$

(C) $\sqrt{(x^2+y^2)}$ (D) $\sqrt{(i^2+j^2+k^2)}$

Q17. Arithmetic Mean between 1/3 and 1/4 is _____.

(A) 7/24 (B) 14/12

(C) 7/12 (D) None of these

Q18. If 3,9,27 ____ is G.P then a6 is _____.

(A) 3^5 (B) 3^6

(C) 3^{-5} (D) 3^{-6}

Q19. General Term of $1.2^2+2.3^2+3.4^2+$____ is:

(A) n(n+1) (B) $n^2(n+1)$

(C) $(n+1)^2$ (D) $n(n+1)^2$

Q20. $(n-1)!/(n+1)!$ = _____.

(A) 1/(n+1) (B) 1/(n(n+1))

(C) (n+1) (D) n(n+1)

SECTION-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 50

Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

Q2. Attempt any TEN parts. Each part carries FIVE marks.

i) Separate into real and imaginary parts $(5+2i)^{-1}/(2-3i)$

ii) Solve the quadratic equation $Z^2 - 6Z + 13 = 0$

iii) Find the rank of [[1,0,-2],[2,2,1],[-1,2,3]]

iv) If a = 3i − j − k, b = −2i + 4j − 3k, c = i + 2j − k. Then find a unit vector parallel to 3a + 2b + 4c?

v) Insert three arithmetic means between 6 and 41?

vi) Find the sum of the n terms of the series $1.2^2 + 2.3^2 + 3.4^2 +$ _____.

vii) Find 'n' if ${}^{2n}C3 : {}^{n}C2 = 36 : 3$

viii) Find the middle term in the expansion of $(a/x + x/a)^8$

ix) Find the inverse of f(x) = (2x−1)/(x+1)

x) Prove that $(1+\cos 2\theta)/\sin 2\theta = \cot\theta$

xi) Find Area of Triangle ABC in which a = 7, b = 8, c = 2 are measures of its sides.

xii) Find the exact value of $\sin \cos^{-1}(\sqrt{3}/2)$

xiii) The second term of H.P is 1/2 and the fifth term is −1/4. Find the 12th term.

SECTION-C

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 30

Note: Note: Attempt any THREE questions. Each question carries equal marks.

Q3.

(a) Use Cramer's rule to solve the system x − 2y + z = 2, 2x + 2y − z = 4, x + y = 0?

(b) Let P = i − 3j + 2k, q = i + j, r = 2i + 2j − 4k then find |p − q + r|

Q4.

(a) Convert for x if 1.63 to a common fraction.

(b) Prove that $(\sin\alpha + \sin 9\alpha)/(\cos\alpha + \cos 9\alpha) = \tan 5\alpha$

Q5.

(a) Prove by Mathematical Induction $1^2 + 2^2 + 3^2 + ____ + n^2 = n(n + 1)(2n + 1) / 6$

(b) Find the area of inscribed circle of the triangle with measures of the sides 55m, 25m, 70m.

Q6.

(a) Show that $2\tan^{-1}A = \tan^{-1}(2A/(1 - A^2))$

(b) Find the value of 'n' when $n(n!)/(n-5)! = 12(n!)/(n-4)!$